

MACHINE LEARNING PROJECT

How I Taught a Machine to Hunt Pulsar Stars

A visual showcase of a **Non-Linear Support Vector Machine (SVM)** trained on NASA's HTRU2 deep-space radio survey to classify cosmic signals as **Pulsar Stars** or background noise — with **GridSearchCV** **hyperparameter tuning** and **PCA decision boundary visualization**.

17,898

Radio Signals

8

Features

**RBF
SVM**

Algorithm

~98%

Accuracy

0.954

ROC AUC

What Does This Model Actually Do?

Every second, radio telescopes record thousands of deep-space signals. The vast majority are just **Radio Frequency Interference (RFI)** — cosmic noise. Hidden inside that noise are incredibly rare, rapidly rotating neutron stars called **Pulsars**.

This model was trained on **17,898 real signal recordings** from NASA's HTRU2 survey. Each signal has **8 statistical features** extracted from its pulse shape. It predicts in milliseconds:



Class 0

Cosmic Noise
(90.8%)



Class 1

Pulsar Star
(9.2%)



The Training Data

17,898 signals split 80/20. Stratified split preserves the 9.2% pulsar ratio in both sets — critical for imbalanced data.



The Challenge: Class Imbalance

For every 1 real Pulsar there are ~10 noise signals. A naive model could hit 90.8% accuracy by always predicting "noise" — so we optimize for F1-Score, not accuracy.



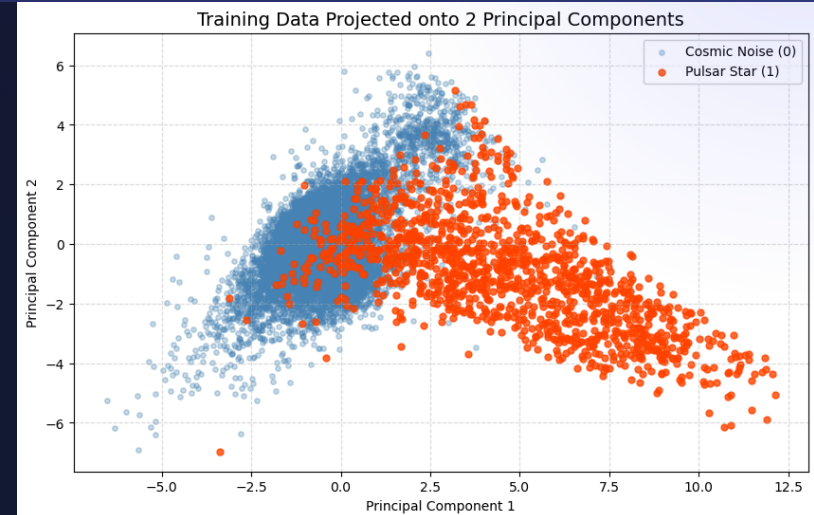
The Output

Given 8 measurements of a new signal: **0 (Noise)** or **1 (Pulsar Star)**.

The 8 Features of a Deep-Space Signal

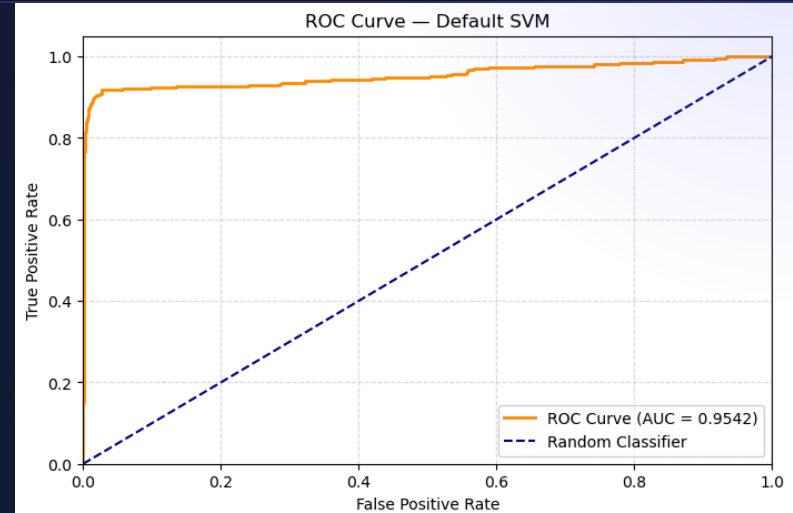
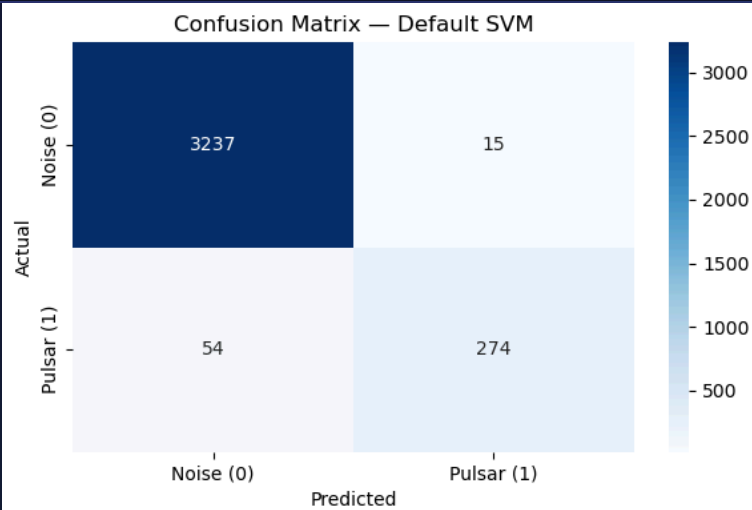
4 stats from the **Integrated Pulse Profile** and 4 from the **DM-SNR Curve**. No missing values.

FEATURE	SOURCE	DESCRIPTION
mean_profile	Profile	Average of pulse shape
std_profile	Profile	Spread / variability
kurtosis_profile	Profile	Sharpness of the peak
skewness_profile	Profile	Asymmetry of shape
mean_dmsnr	DM-SNR	Average signal-to-noise ratio
std_dmsnr	DM-SNR	Spread of SNR curve
kurtosis_dmsnr	DM-SNR	Sharpness of SNR peak
skewness_dmsnr	DM-SNR	Asymmetry of SNR curve
target	Label	0 = Noise 1 = Pulsar ★



Why PCA? 8 features can't be plotted directly. PCA compresses them to 2 axes of maximum variance — confirming Pulsars and Noise form separable clusters before training.

Default SVM — Confusion Matrix & ROC Curve



The Non-Linear Boundary the SVM Drew



What is the boundary?

The curved black line is the decision boundary in 2D PCA space. Points on one side = Noise, other side = Pulsar. Unlike Logistic Regression, this boundary is non-linear — it curves and bends to fit complex patterns.



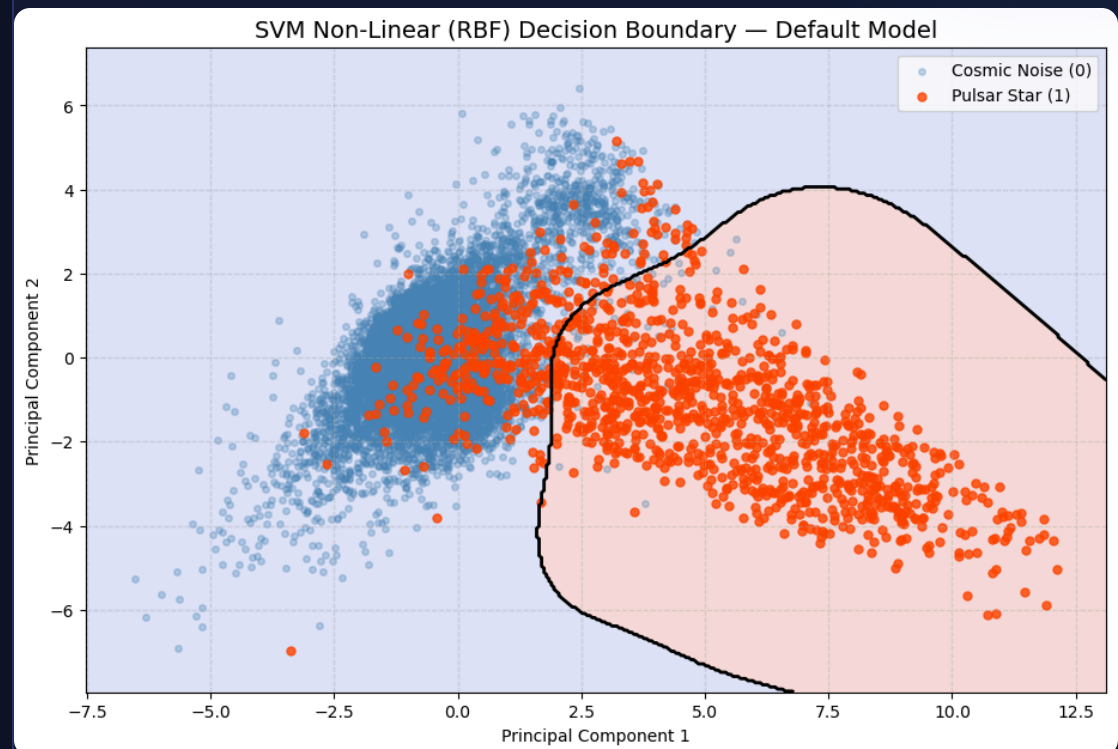
Why RBF Kernel?

The RBF (Gaussian) kernel maps data into infinite-dimensional space where a linear separator exists. Controlled by C (margin) and γ (influence radius).



Note on 2D vs 8D

The real model trains on all 8 features. This plot uses a 2-component PCA projection purely for visualization. The actual 8D boundary is far more precise.



What We Changed — GridSearchCV Tuning

The default SVM uses `C=1`, `gamma='scale'` — just initial guesses.

GridSearchCV systematically tests every combination using 5-fold cross-validation scored by **F1-Score** (not accuracy — because of class imbalance).

C
`0.1/1/10/100`

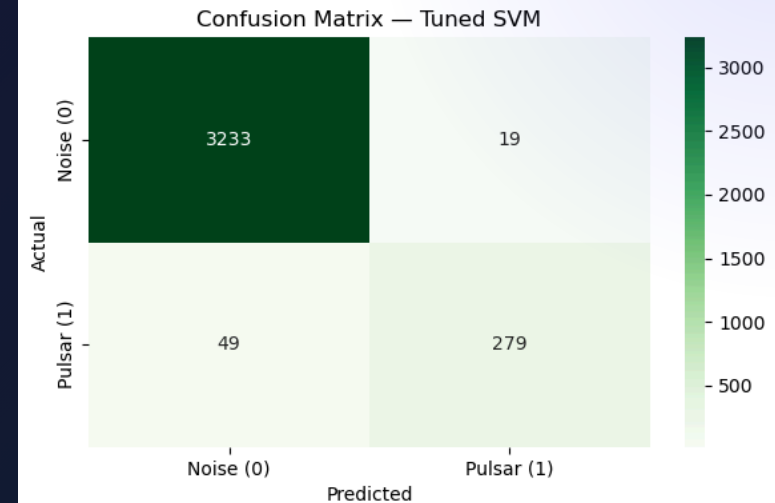
GAMMA
`scale/auto/0.01/0.1`

CLASS WEIGHT
`None/balanced`

`class_weight='balanced'` auto-adjusts penalties for the rare Pulsar class, making the model more aggressive at catching Pulsars — reducing False Negatives. A justified trade-off in astronomical discovery.

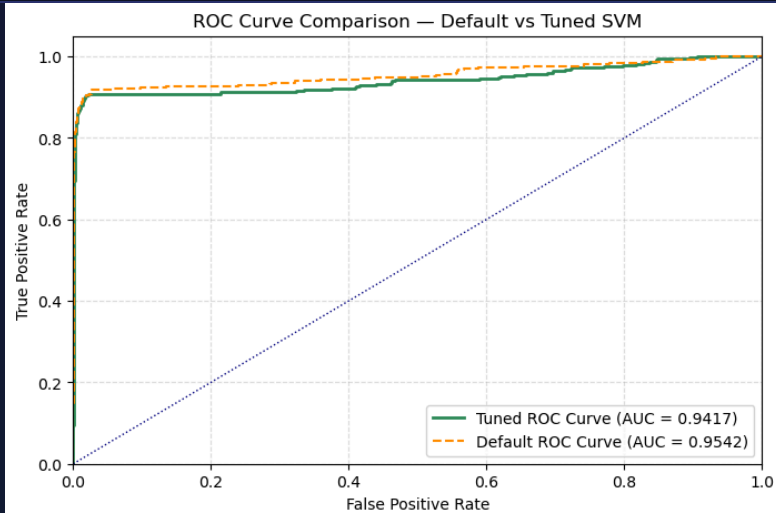
BEST PARAMETERS FOUND

```
C=1, gamma='auto', class_weight='balanced'
```

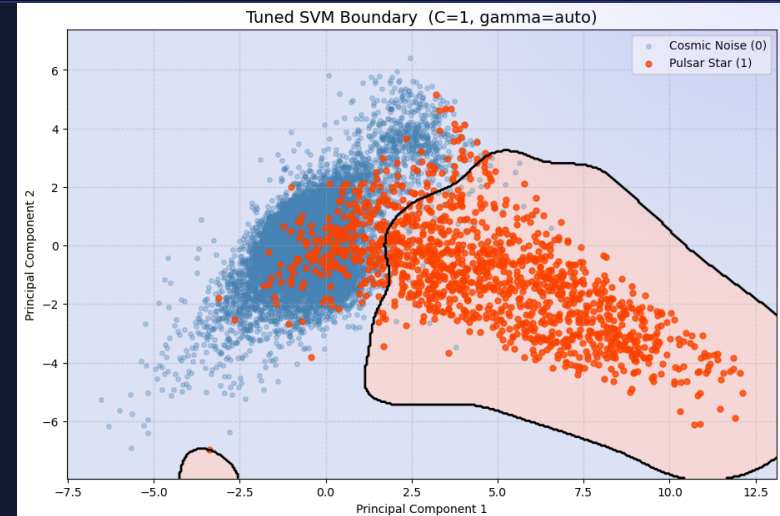


Notice fewer **False Negatives** (missed Pulsars) vs. the default model — the primary goal of tuning.

Tuned SVM — ROC Comparison & Decision Boundary



The **green tuned curve** vs **orange default curve**. Note: tuning optimized F1-Score (not AUC), improving Pulsar recall at a minor precision trade-off.



With `gamma='auto'`, the boundary adapts more tightly to the Pulsar cluster, correctly capturing more **True Positives** in the minority class.